

Choosing and Building the Right Forecast

Module 5, Part 1 (December 4)

QBA 5330

Where We're Headed Today

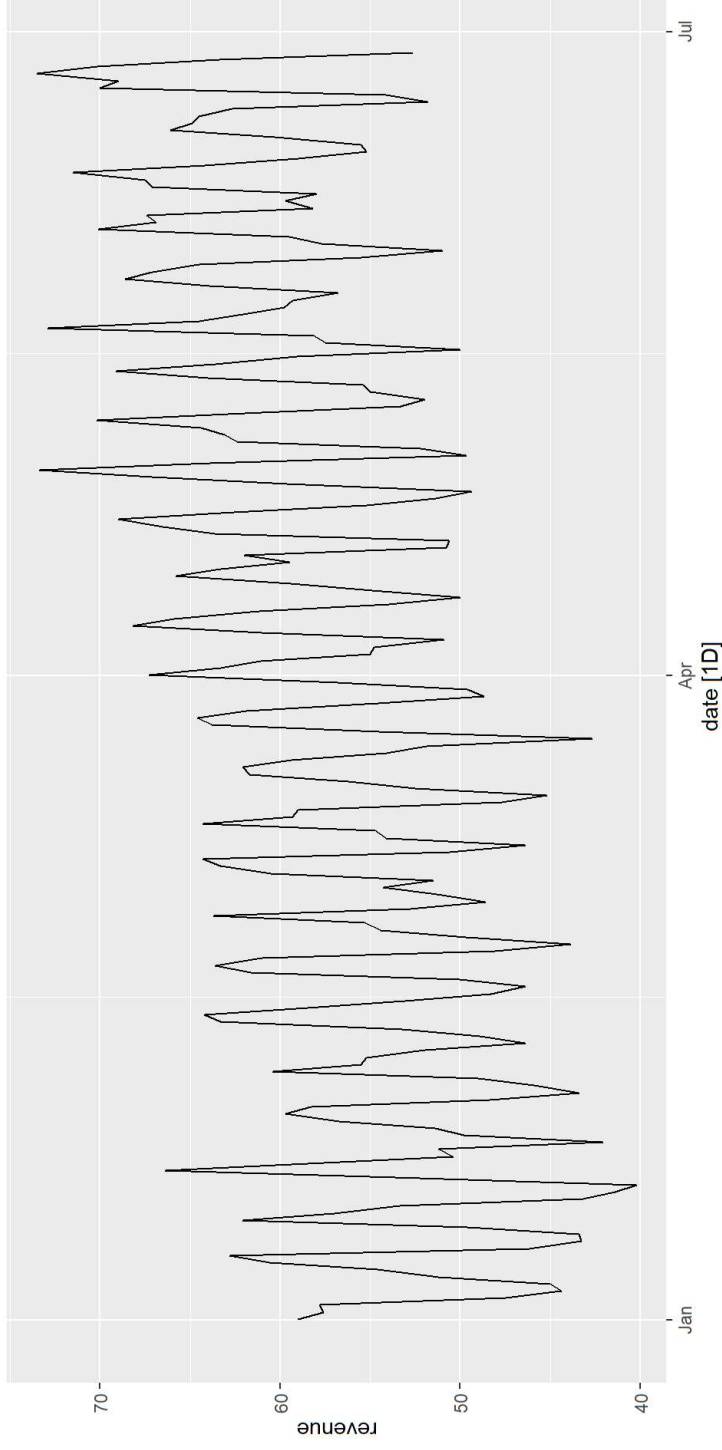
- Part 1: Choosing and building the right forecast
- Part 2: Forecasting Ceres's growth
- Part 3: Growth vs. cash flow — preparing for the bank

Recall: The Full Toolkit

- Naïve / seasonal naïve
- Moving averages
- Exponential smoothing (SES, Holt's, Holt-Winters)
- Regression-based forecasting (TSLM)
- *(All already in your reading – today applies it, fast.)*

Live Coding: Back to Back

```
1 library(tidyverse)
2 library(fpp3)
3
4 coffee <- read_csv("data/coffee_shop_daily.csv") |> as_tsibble(index = date)
5 coffee |> autoplot(revenue)
```



Naïve vs. Seasonal Naïve

```
1 coffee |> model(naive = NAIVE(revenue), snaive = SNAIVE(revenue)) |>  
2   forecast(h = "1 day") |> hilo(level = 95)  
  
# A tibble: 2 x 5 [1D]  
# Key:      .model [2]  
  .model date  
  <chr> <date>  
1 naive 2024-06-29  
2 snaive 2024-06-29  
# i 3 more variables: revenue <dist>, .mean <dbl>, `95%` <hilo>
```

Holt-Winters

```
1 coffee |> model(hw = ETS(revenue ~ error("A") + trend("A") + season("A")))|>  
2   report()
```

Series: revenue

Model: ETS(A,A,A)

Smoothing parameters:

alpha = 0.04102127

beta = 0.0001000002

gamma = 0.0001002058

Initial states:

l[0]	b[0]	s[0]	s[-1]	s[-2]	s[-3]	s[-4]	s[-5]
52.43567	0.07036101	-0.2289435	-5.763454	-7.679007	-3.839494	3.35794	7.180701

s[-6]
6.972257

sigma^2: 10.0995

Activity: Which Method Would You Use?

No calculation – just diagnosis.

Debrief

- Scenario 3: what would you actually fit?

Later: A Real Company

- Ceres Gardening Company's CEO has two things on his to-do list:
 - A motivational speech for the sales team
 - A meeting with the **bank** about growth and cash flow
- **We're helping with the second one.**

Before the Break

- Ceres's own financial reporting is **annual only** – 5–6 data points
- Nowhere near enough for moving averages or seasonality
- **Next: a supplementary monthly series, then back to Ceres's real numbers.**